

화학수학

열여덟 번째 수업

군론 / 확률과 통계

Full Bag of New Terms

1. **Special matrices:** identity matrix, zero matrix
2. **Partner of a matrix:** inverse, transpose, Hermitian conjugate
3. **Properties of a matrix:** singular, symmetric, Hermitian, orthogonal, unitary
4. **Derived quantities from a matrix:** trace, determinant

Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows $\rightarrow$ columns, columns $\rightarrow$ rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$(\tilde{\mathbf{A}})_{ij} = \tilde{a}_{ij} = a_{ji}$$

Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

(에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$(\mathbf{A}^\dagger)_{ij} = a_{ji}^*$$

Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse
- Symmetric matrix (대칭행렬): $\mathbf{A} = \tilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬): $\mathbf{A} = \mathbf{A}^\dagger$
- Orthogonal matrix (직교행렬): $\mathbf{A}^{-1} = \tilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬): $\mathbf{A}^{-1} = \mathbf{A}^\dagger$

Trace of a Square Matrix (대각합)

$$\text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \text{tr } \mathbf{A} = a_{11} + a_{22} + a_{33}$$

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Jacobian calculation
- Determinant of a singular matrix
- Matrix eigenvalue problem

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

Today's Class

- Groups
 - Point group
 - Representation of a point group
 - Characters
 - Irreducible representations
 - Character tables
- Errors and probability theory

Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A)$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds: $A(BC) = (AB)C$

Example

group of $\{1, i, -1, -i\}$

- Multiplication table

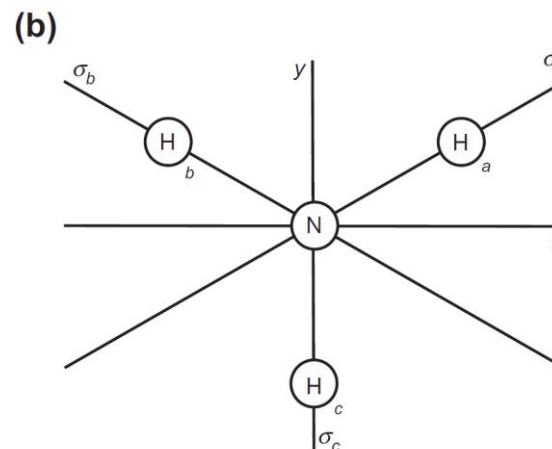
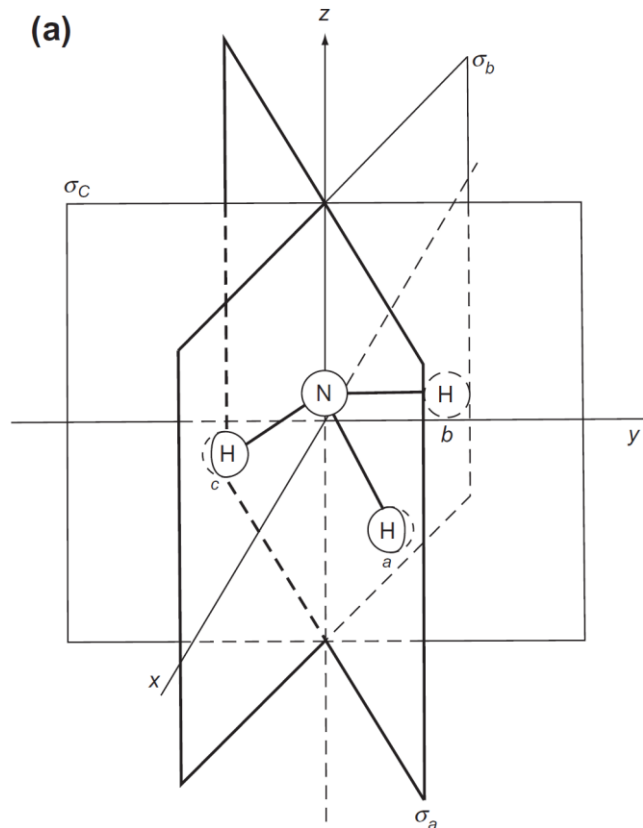
	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

Identity = 1

Point Group (점군)

Formed by a set of point symmetry operators
belonging to a symmetrical object

Example: Ammonia (NH₃)

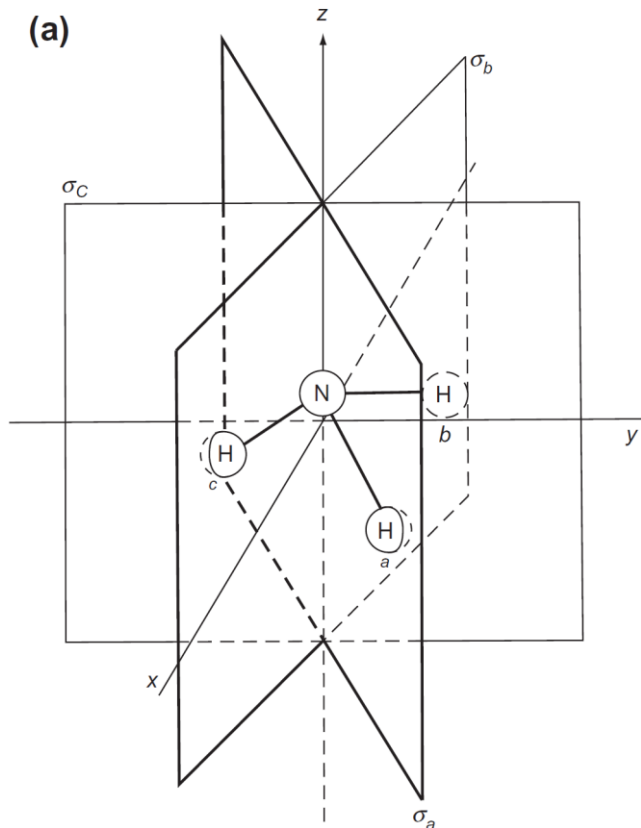


Symmetry elements: $C_3, 3 \times \sigma_v$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

Example: Ammonia (NH₃)



Symmetry elements: $C_3, 3 \times \sigma_v$

Symmetry operators:

$$\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c$$

Multiplication table:

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

Schoenflies Symbol

Each point group is assigned a symbol

Symbol	Symmetry elements
C_1	E
C_s	E, σ
C_i	E, i
C_n	E, C_n
S_n	E, S_n
C_{nv}	$E, C_n, n \times \sigma_v$
C_{nh}	E, C_n, σ_h
D_n	$E, C_n, n \times C_2$
D_{nh}	$E, C_n, n \times C_2, \sigma_h$
D_{nd}	$E, C_n, n \times C_2, n \times \sigma_v$
T_d	$E, 6 \times \sigma_d, 4 \times C_3, 3 \times S_4$
O_h	$E, 6 \times \sigma_d, 3 \times \sigma_h, 6 \times C_2, 4 \times C_3, 3 \times C_4, 3 \times S_4, 4 \times S_6$

Representation of a Point Group

Symmetry operator $\leftrightarrow$ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

Identity Symmetry Operator

$$\hat{E} \leftrightarrow \mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) ,
determine a_{ij}

Example 13.20

Find the matrix that represents $\hat{C}_{n(z)}$.

The operation transforms (x_1, y_1, z_1) to (x_2, y_2, z_2) as follows:

$$x_2 = \left(\cos \frac{2\pi}{n} \right) x_1 - \left(\sin \frac{2\pi}{n} \right) y_1$$

$$y_2 = \left(\sin \frac{2\pi}{n} \right) x_1 + \left(\cos \frac{2\pi}{n} \right) y_1$$

$$z_2 = z_1$$

Hence, the representation matrix is

$$\hat{C}_{n(z)} \leftrightarrow \begin{bmatrix} \cos(2\pi/n) & -\sin(2\pi/n) & 0 \\ \sin(2\pi/n) & \cos(2\pi/n) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Representation of the C_{3v} Group

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

Example 13.21

By matrix multiplication, show that $\mathbf{AB} = \mathbf{E}$.

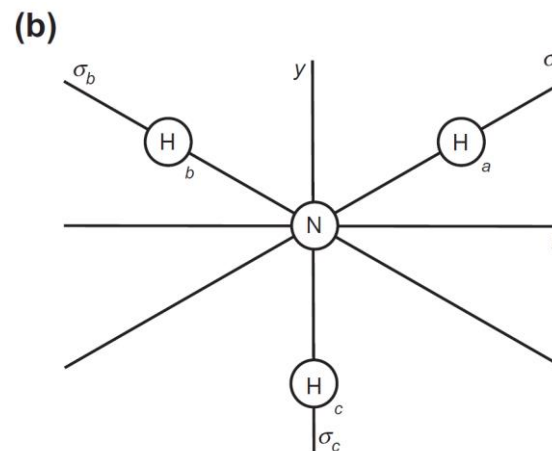
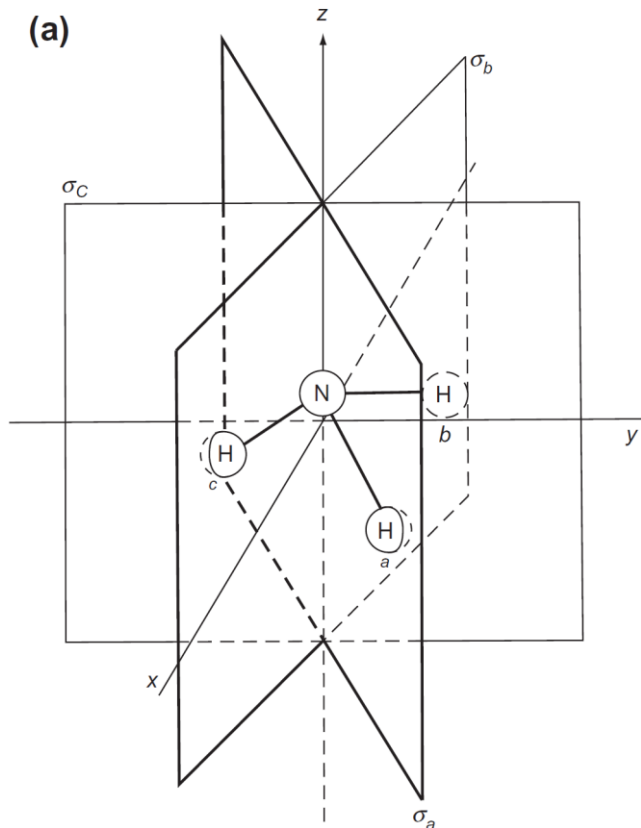
Since

$$\mathbf{A} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

matrix multiplication gives

$$\mathbf{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}$$

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

Similarity Transformation (답음 변환)

- Two different representations **P** and **Q** in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when **X** is some square matrix

- Since $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \text{tr}(\mathbf{XX}^{-1}\mathbf{Q}) = \text{tr}(\mathbf{Q})$$

Character (지표)

- **Character** of a symmetry operator = **trace** of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Character of } \hat{C}_3 = \text{tr } \mathbf{A} = (-1/2) + (-1/2) + 1 = 0$$

Our Representation of C_{3v} Group

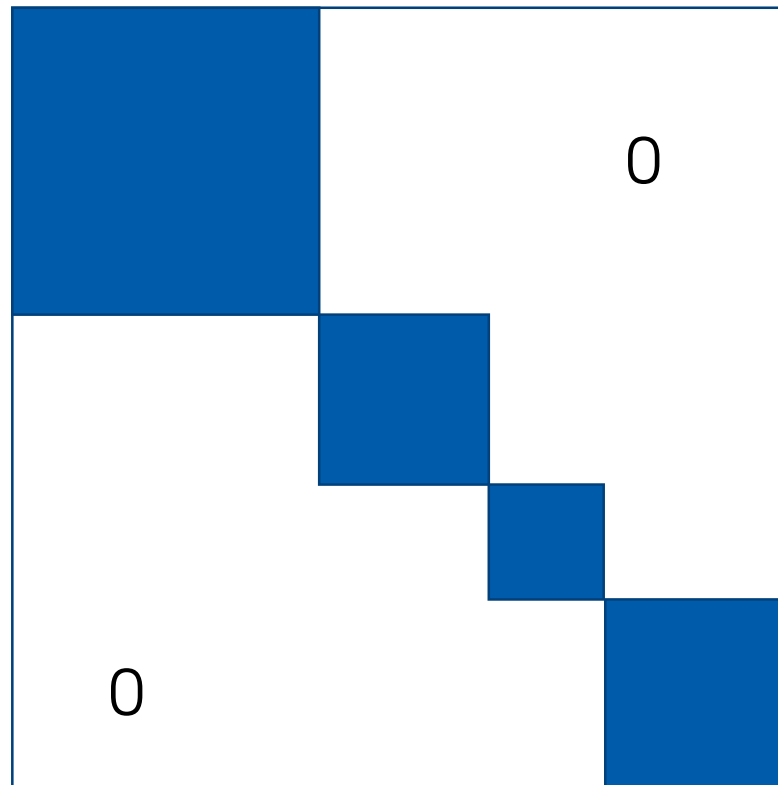
$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
3	0	0	1	1	1

Block-Diagonal Matrix (블록대각행렬)



Reducible Representation (가약 표현)

All matrices are **block-diagonal**, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{F}$$

Block Submatrices

- Block submatrices obey the same multiplication table of the group of symmetry operators
- Corresponding block submatrices can form a new representation

Example: 2 by 2 Blocks

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Example: 1 by 1 Blocks

$$\begin{aligned}\hat{E} &\leftrightarrow 1, & \hat{C}_3 &\leftrightarrow 1, & \hat{C}_3^2 &\leftrightarrow 1, \\ \hat{\sigma}_a &\leftrightarrow 1, & \hat{\sigma}_b &\leftrightarrow 1, & \hat{\sigma}_c &\leftrightarrow 1\end{aligned}$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal
by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as **irrep**

Examples of C_{3v}

- 2 by 2 blocks: one irrep

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- 1 by 1 blocks: another irrep

$$\hat{E} \leftrightarrow 1, \quad \hat{C}_3 \leftrightarrow 1, \quad \hat{C}_3^2 \leftrightarrow 1, \quad \hat{\sigma}_a \leftrightarrow 1, \quad \hat{\sigma}_b \leftrightarrow 1, \quad \hat{\sigma}_c \leftrightarrow 1$$

Characters of Irreps

Class:	E	$2C_3$		$3\sigma_v$		
Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
2 by 2	2	-1	-1	0	0	0
1 by 1	1	1	1	1	1	1

Class is constituted of symmetry operations
that can be converted by similarity transformations

Character Table (지표표)

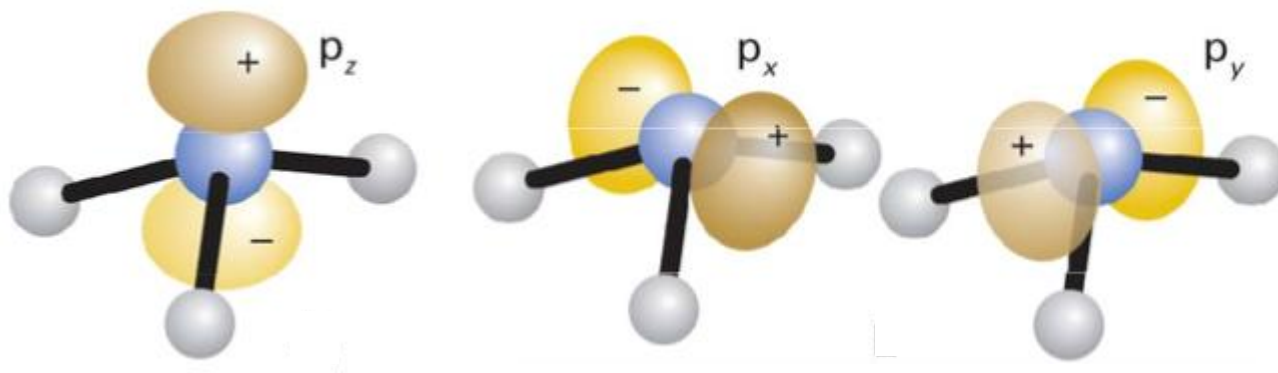
- Using group theory theorems, we can find all possible irreps and collect their characters → **character table**
- e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

Application of Character Table

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



(Image: https://commons.wikimedia.org/wiki/File:Orbitals_of_NH3.png)

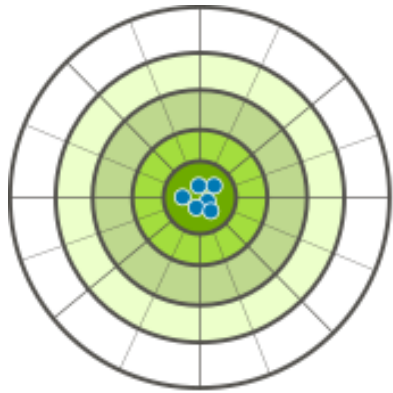
15. Probability, Statistics, and Experimental Errors

(15장 확률, 통계, 실험오차)

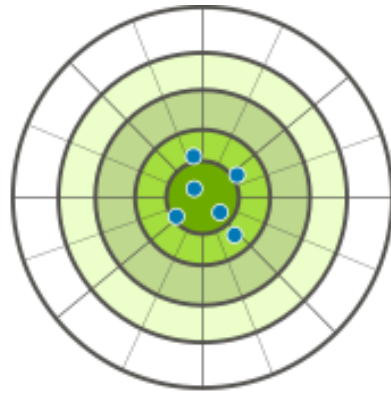
Errors in Measurement

- **Systematic errors** (계통오차)
 - same direction and same magnitude for repeated measurements
 - due to limitations in the apparatus
- **Random errors** (우연오차)
 - random direction and magnitude
 - due to limitations in the apparatus, limitations in the techniques, unavoidable human error

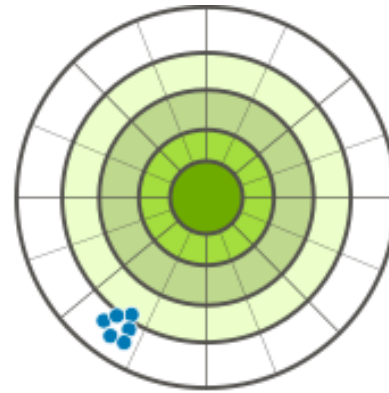
Precision (정밀도) and Accuracy (정확도)



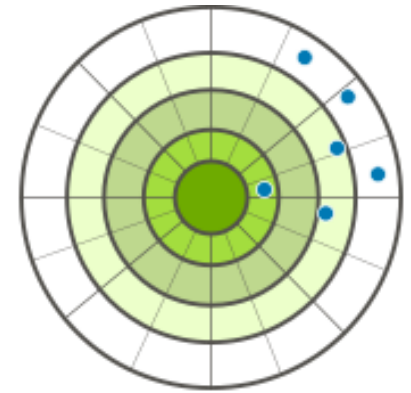
accurate
precise



accurate
not precise



not accurate
precise



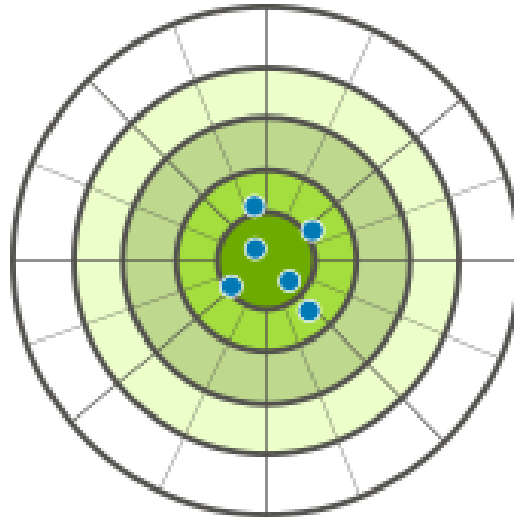
not accurate
not precise

random errors

systematic errors

Repeated Measurements

sampling from a population of imaginary measurements with random errors



(Image: <https://openclipart.org/detail/191649/accuracy-vs-precision>)

Probability Theory

study of random events

- **Classical systems:** unknown information
- **Quantum systems:** intrinsic randomness
- **Population:** large set of objects or numbers
 - Classical systems: imaginary set of infinitely many repetitions of a given measurement
 - Quantum systems: imaginary set of infinitely many replicas of the physical system

Probability Distribution (확률분포)

mathematical function that gives
the probabilities of occurrence of
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):
possible outcomes have continuous values